

Smoothed Particle Hydrodynamics in one spatial dimension

- we aim at writing a 1D SPH code that solves the following equations of hydrodynamics for a set of N particles

$$\frac{dx_i}{dt} = v_i$$

$$\frac{dv_i}{dt} = - \sum_j m_j \left(\frac{P_j}{\rho_j^2} + \frac{P_i}{\rho_i^2} \right) \frac{dW_{ij}}{dx_i}$$

$$\frac{d\rho_i}{dt} = \sum_j m_j (v_i - v_j) \frac{dW_{ij}}{dx_i}$$

$$\frac{de_i}{dt} = \frac{1}{2} \sum_j m_j \left(\frac{P_j}{\rho_j^2} + \frac{P_i}{\rho_i^2} \right) (v_i - v_j) \frac{dW_{ij}}{dx_i}$$

$$P = (\gamma - 1) \rho e$$

$$W_{ij} = W(|x_i - x_j|, h_i)$$

$$\frac{dW_{ij}}{dx_i} = \frac{dW(r, h_i)}{dr} \frac{dr}{dx_i}$$

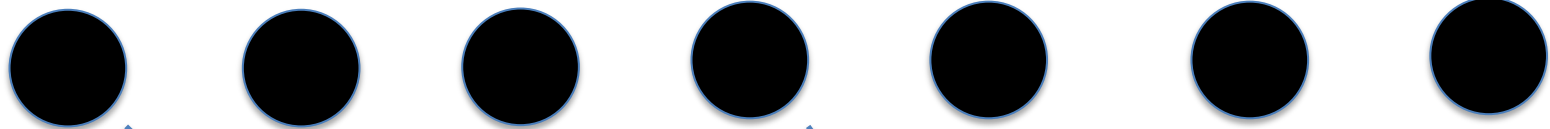
$$r = |x_i - x_j|$$

$$W(r, h) = \frac{2}{3h} \begin{cases} 1 - \frac{3}{2} \left(\frac{r}{h} \right)^2 + \frac{3}{4} \left(\frac{r}{h} \right)^3 & 0 \leq \frac{r}{h} \leq 1 \\ \frac{1}{4} \left(2 - \left(\frac{r}{h} \right) \right)^3 & 1 \leq \frac{r}{h} \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Smoothed Particle Hydrodynamics in one spatial dimension: blastwave

▪ we want to simulate a supernova blast wave:

- the initial conditions are given by placing those N particles down on the interval $[0,1]$ using a spacing $dx=1/(N-1)$
- set the thermal energies of all particles to $e=1e-5$, only the central particle gets $e=1$ hence triggering a blast wave
- use an adiabatic coefficient of $\gamma=7/5$ for this test

 $x=0$ $x=1$ 

m	1
x	$[0,1]$
v	0
rho	m/dx
e	$1e-5$

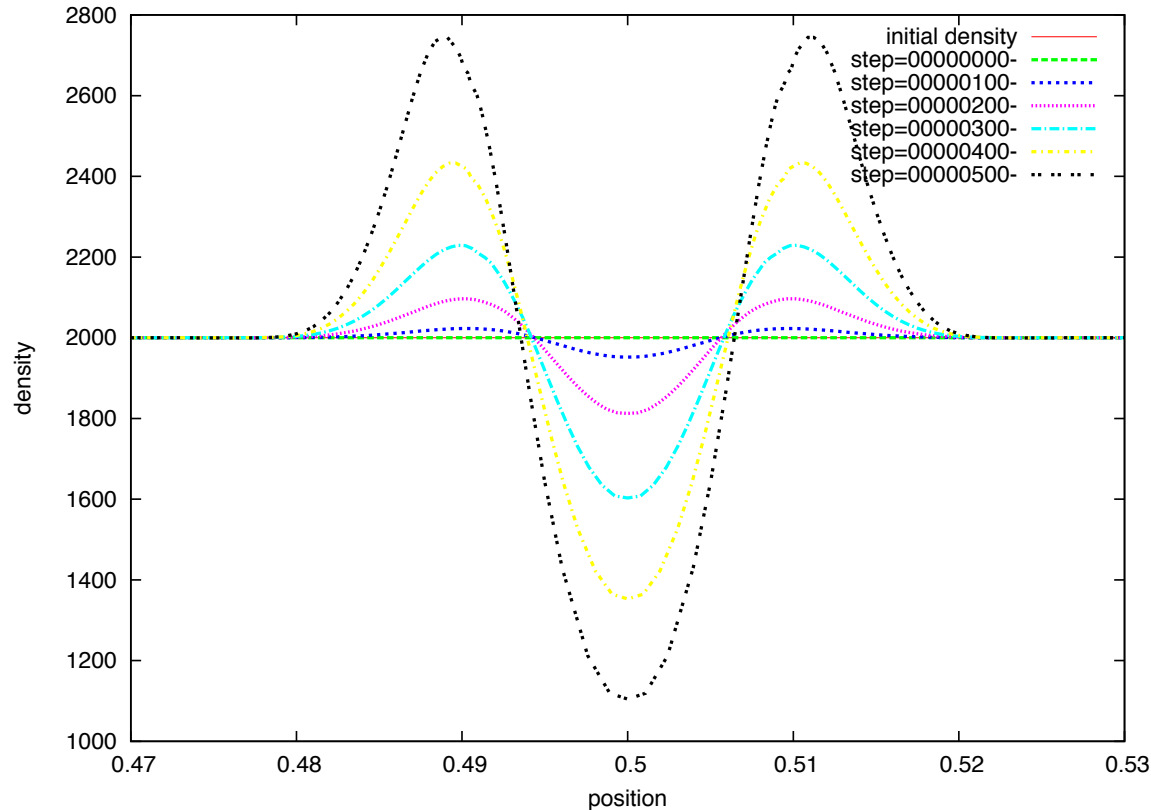
only for the central particle set $e=1$

Smoothed Particle Hydrodynamics in one spatial dimension: blastwave

▪ for the following setup

- NPART = 2000 (number of gas particles)
- NSTEPS = 500 (number of integration steps)
- NSPH = 25 (number of SPH neighbour particles used with the solver)
- Tend = 0.05 (end-time of the integration, Tstart=0)

you should be able to generate a plot similar to this one.



1.loop (parallel): calculate the coefficients for each particle i

$$v_coeff_i = - \sum_j m_j \left(\frac{P_j}{\rho_j^2} + \frac{P_i}{\rho_i^2} \right) \frac{dW_{ij}}{dx_i}$$

$$rho_coeff_i = \sum_j m_j (v_i - v_j) \frac{dW_{ij}}{dx_i}$$

$$e_coeff_i = \frac{1}{2} \sum_j m_j \left(\frac{P_j}{\rho_j^2} + \frac{P_i}{\rho_i^2} \right) (v_i - v_j) \frac{dW_{ij}}{dx_i}$$

Note: W_{ij} (and hence dW_{ij}/dx_i) is zero for $|x_i - x_j| > 2h_i$

2.loop (parallel): use coefficients to perform integration step for each particle i

$$x_i \mapsto x_i + v_i \Delta t$$

$$v_i \mapsto v_i + v_coeff_i \Delta t$$

$$\rho_i \mapsto \rho_i + \rho_coeff_i \Delta t$$

$$e_i \mapsto e_i + e_coeff_i \Delta t$$

each particle is a structure containing the actual quantities as well as the coefficients

```
struct particle {  
  
    double x;  
    double v;  
    double rho;  
    double e;  
    double h;  
  
    double v_coeff;  
    double rho_coeff;  
    double e_coeff;  
  
};
```

some pseudo C-code for main routine

```
main()
{
    initial_conditions();

    while-loop for the time stepping()
    {
        set_h();
        getSPHcoefficient();
        performIntegration();
        updateTimeCounter();

        WriteOutputFile();
    }
}
```

some pseudo C-code for main routine

```
main()
{
    initial_conditions();

    while-loop for the time stepping()
    {
        set_h();  how to set the individual h-values?
        getSPHcoefficient();
        performIntegration();
        updateTimeCounter();

        WriteOutputFile();
    }
}
```

how to find the NSPH nearest neighbours of particle i ????

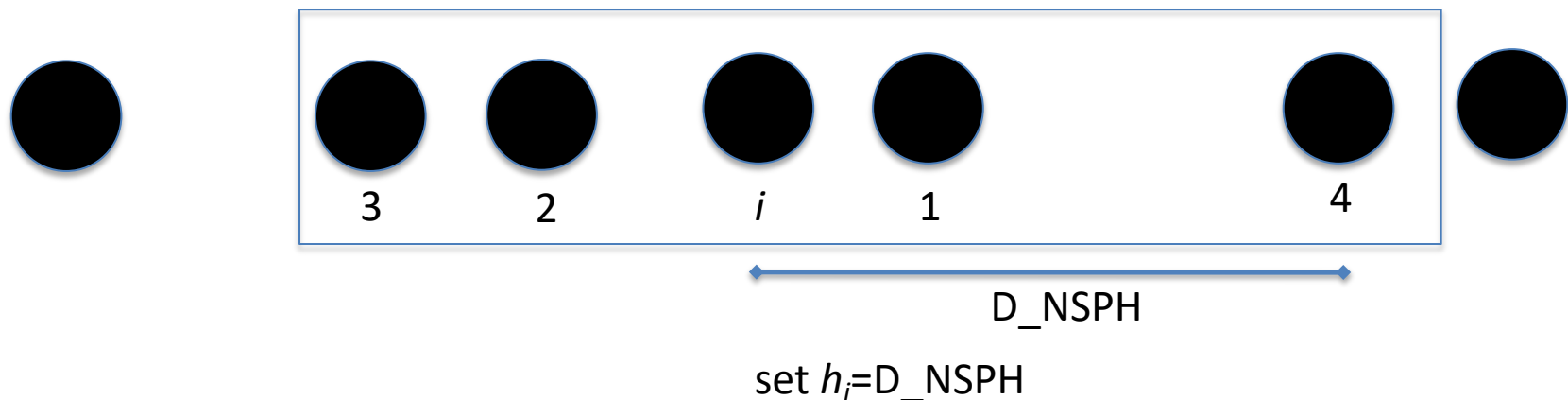
1. calculate the distance of every particle j to particle i
2. sort the particles with respect to distance
3. only consider the first NSPH particles in this ordered list
4. set h to be the distance to particle NPSH

how to find the NSPH nearest neighbours of particle i ????

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example: NSPH=4

only consider the 4 nearest neighbours to particle i

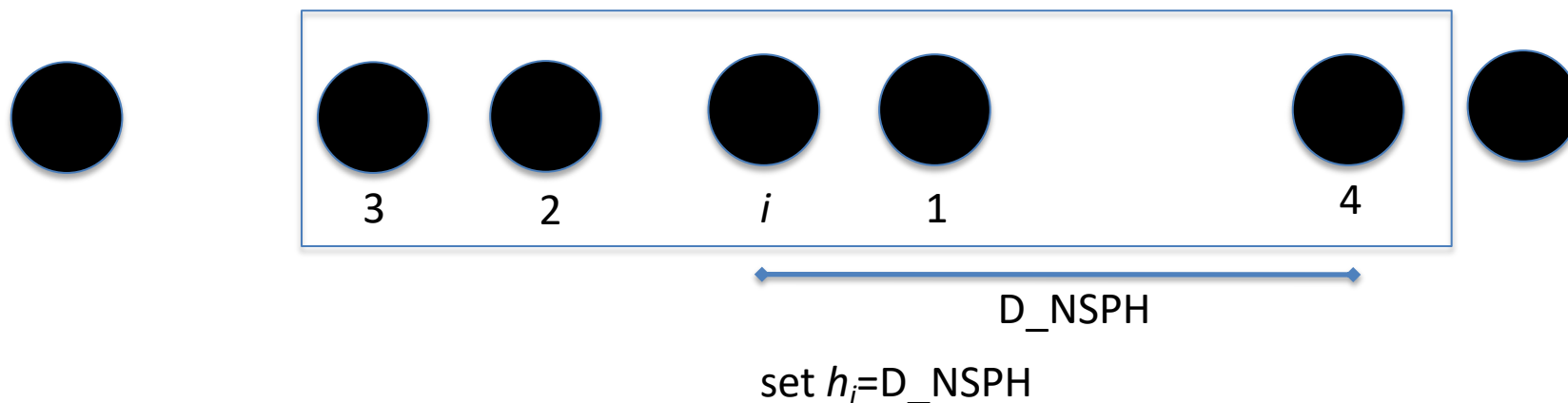


how to find the NSPH nearest neighbours of particle i ????

1. calculate the distance of every particle j to particle i
2. **sort the particles with respect to distance????**
3. only consider the first NSPH particles in this ordered list
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possible improvements

- symmetrize kernel: $W_{ij} \rightarrow \frac{1}{2} \left(W(|x_i - x_j|, h_i) + W(|x_i - x_j|, h_j) \right)$
- smooth initial explosion
- periodic boundary conditions
- 2nd order Runge-Kutta integration
- Apply your tree code for neighbour finding